

```

1  import os
2  import ast
3  from pathlib import Path
4
5  class ConfigReader:
6      def __init__(self, config_path):
7          self.config = {}
8          self.config_path = config_path
9          self._load_config()
10
11     def _load_config(self):
12         """Load configuration from properties file."""
13         with open(self.config_path, 'r') as f:
14             for line in f:
15                 line = line.strip()
16                 if line and not line.startswith('#'):
17                     key, value = line.split('=', 1)
18                     self.config[key.strip()] = value.strip()
19
20     def get(self, key, default=None):
21         """Get configuration value with optional default."""
22         return self.config.get(key, default)
23
24     def get_int(self, key, default=None):
25         """Get integer configuration value."""
26         value = self.get(key, default)
27         return int(value) if value is not None else default
28
29     def get_float(self, key, default=None):
30         """Get float configuration value."""
31         value = self.get(key, default)
32         return float(value) if value is not None else default
33
34     def get_bool(self, key, default=None):
35         """Get boolean configuration value."""
36         value = self.get(key, default)
37         if value is None:
38             return default
39         return value.lower() == 'true'
40
41     def get_list(self, key, default=None, type_func=str):
42         """Get list configuration value."""
43         value = self.get(key, default)
44         if value is None:
45             return default
46         return [type_func(x.strip()) for x in value.split(',')]
47
48     def get_tuple(self, key, default=None, type_func=int):
49         """Get tuple configuration value."""
50         value = self.get(key, default)
51         if value is None:
52             return default
53         return tuple(type_func(x.strip()) for x in value.split(','))
54
55     def get_path(self, key, default=None):
56         """Get path configuration value."""
57         value = self.get(key, default)
58         if value is None:
59             return default
60         return Path(value)

```